Research Report

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3D–2D Medical Image Registration Using Deep Learning

SUBMITTED BY

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Chapter 1

INTRODUCTION

Medical image registration is the process of estimating the spatial transformation that aligns two images by bringing corresponding anatomical structures into spatial correspondence. This process is crucial for image-guided procedures and surgeries wherein the pre-operative anatomical data have to be related to the images captured during the procedure. In the case of 3D– 2D registration, a 3D volume is matched against a 2D image; hence, this problem is inherently different from the standard volumetric registration[1, 2].

The study focuses on pre-operative spinal magnetic resonance imaging (MRI) with a slice thickness of 3 mm and the corresponding intra-operative two-dimensional X-ray images. MRI encodes volumetric anatomical information while an X-ray records anatomy projected onto a detector from a particular view. The two modalities have different image formation processes. As such, MRI intensities cannot be directly interpreted in terms of X-ray attenuation and are thus generally not suitable for direct raw-intensity matching. Vertebral endplates, contours and landmarks provide a more meaningful basis for cross-modal correspondence[3–5].

Prior work encompasses direct prediction of transformation parameters, learned iterative refinement, differentiable X-ray rendering, semantic feature extraction, and feature-based MRI-to-X-ray registration. Several studies have forward-projected vertebral features segmented from MRI to construct simulated X-ray representations; others have aligned pre-operative MRI with intra-operative radiographs through anatomical features and geometric optimization[3–5]. Simulated projection and feature-based MRI-to-X-ray registration are thus established approaches, rather than novel contributions within themselves. The current study asks whether a learned feature representation improves upon a conventional feature-based pipeline given the available spinal MRI and intra-operative X-ray data.

The proposed framework incorporates learned features with explicit imaging geometry, which will extract vertebral features from MRI and X-ray, project the MRI-derived anatomical representation to 2D using an appropriate projection model, match corre-

sponding features, and compute a rigid transformation. As MRI is not a CT volume, simulated views in this framework will be based on segmented or otherwise extracted anatomy, as opposed to uncalibrated MRI intensities being treated as X-ray attenuation values. We will choose the feature encoder and optimization strategy after assessing the dataset size, annotations, and imaging geometry.

Chapter 2

Research Problem Statement

The problem at hand is to estimate a rigid spatial transformation that aligns a pre-operative three-dimensional spinal MRI volume with an intra-operative two-dimensional X-ray image. The rigid transformation has six degrees of freedom, namely, three translations and three rotations. The main difficulties to be overcome are loss of depth information in projection, dissimilarity of appearance between MRI and X-ray images, ambiguity in the appearance of vertebrae in the image, and sensitivity to initial pose and projection geometry[1, 3].

The main components of this problem have been studied before. MR-level check used projections of segmented MRI and gradient-orientation matching to align pre-operative MRIs with intra-operative radiographs. Later studies that used features extracted vertebral endplates and simulated projections or geometric optimization for MRI-to-X-ray alignment[3–5]. The present work therefore does not make a novelty claim about the use of MRI, simulated X–rays, vertebral features or rigid 3D–2D registration. It will test a more restricted hypothesis that suggests that learned feature extraction or description improves over a conventional feature based baseline under equivalent geometric and evaluation conditions.

The central research question is: **Can learning-based extraction or description of vertebral features improve rigid feature-based 3D–2D registration between pre-operative spinal MRI and intra-operative X-ray, relative to a conventional anatomical-feature baseline, when evaluated on simulated projections and the available real X-ray data?** Simulated projections will be generated from MRI derived anatomical structures using available or estimated projection geometries. They will allow for controlled pose perturbations and provide ground truth transformations suitable for quantitative testing. The ability to process real clinical intra-operative X-ray images will be evaluated on a separate dataset to avoid conflation of results on simulated projections with those on real images.

The viability of the learning component is determined by how many independent cases are available, and the quality of annotations that can be extracted. For a small dataset,

it may not be viable to train a deep network from scratch, whereas a compact model, transfer learning or a traditional feature-based approach could be more justified. The final method and the extent of contribution will be articulated following the inventory of the dataset and imaging-geometry checks.

Chapter 3

Literature Survey

Deep-learning-based 3D–2D registration encompasses techniques for direct prediction of transformation parameters as well as hybrid approaches that combine learned representations with geometric constraints and iterative refinement. In particular, the present project, MRI-to-radiograph registration and vertebral feature extraction, is relevant to both research areas due to their similar nature of cross-modal alignment.

Miao et al. proposed a hierarchical convolutional neural network (CNN) to regress transformation parameters from digitally reconstructed radiographs (DRRs) and the observed X-ray image. Jaganathan et al. later presented an iterative approach that learns pose updates as well as incorporates a point-to-plane correspondence solver, which jointly estimates learning-based transformations and explicitly performs geometric computations[2, 6]. Related work addresses the synthetic-to-real domain gap through adaptation at feature and pixel space, using differentiable rendering, and learning with patient-specific supervision[7–9].These works commonly use CT or another volumetric representation suitable for X-ray rendering, and hence their performance should not be extrapolated onto spinal MRI case.

De Silva et al. present an MR-LevelCheck method that segments vertebral structures from pre-operative MRI scans, forward projects the segmentations, and aligns them with intra-operative radiographs by optimizing their gradient orientation similarities using covariance matrix adaptation evolution strategy The authors report, based on a study involving 10 patients, a median projection-distance error of 4.3 mm (interquartile range, 2.6 mm) and no registration failures under their protocol[3]. This is indeed a direct evidence that MRI-to-intra-operative-radiograph registration using segmented anatomy was performed even before the current project's undertaking.

The Kiran et al. paper describes a feature-based multimodal framework that makes it possible to extract vertebral endplates from MRI, forward-project them into simulated X-ray views, and optimize binary image similarity with the CMA-ES algorithm. The method was evaluated using simulated anteroposterior (AP) and lateral projections, resulting in 95% success and approximately 60 seconds of computation time per single

view experiment[4]. The study is closely related to the proposed projection-and-matching pipeline, although its main experiments used simulated views rather than a comprehensive evaluation on real intra-operative X-rays.

In a later work, Kiran et al. combine pre-operative MRI images and intra-operative X-ray projections to estimate the vertebral pose and reconstruct the trajectory of the pedicle-screws. They also make use of vertebral endplates, binary feature descriptors and CMA-ES optimisation. The reported MRI slice thickness is 3 mm. projection-distance error below 5 mm, and target-registration error below 2.5 mm in its evaluated setting [5]. This is an especially close methodological and data-level precedent. The provenance of the dataset available for the present project should therefore be cross-referenced against this publication, and any discrepancy in the data, implementation, or research question should be made explicit before a claim of novelty can be made.

Ku et al. present an MRI-to-fluoroscopy pipeline for orthopaedic procedures using deep learning to generate a synthetic CT (sCT) from MRI before DRR generation and 2D–3D registration. Their evaluation included simulation and cadaver studies of the femur and pelvis [10]. Unlike the proposed spinal workflow, none of the works attempt to investigate the projections of MRI-derived anatomical features without relying on an sCT volume as an intermediate

For the spinal anatomy, Qu et al. leverage the graph-based feature aggregation with the transformer modeling for the radiograph-to-CT registration task, whereas Zhao et al. suggest the two-step transformer-based architecture with the semantic attention mechanism, and Shen et al. utilize the semantic vertebral edges as a guiding force for the single-view lumbar registration. Altogether, these three studies confirm the validity of anatomy-aware representation, yet their registration objective primarily focuses on the alignment of CT with the X-ray images, and not the testing of the learned features for the MRI-to-X-ray registration task [11–13]. Unberath et al. provide a broader review of learning-based 2D–3D registration, including geometric modelling, generalization, and verification challenges [1].

3.1 Summary of Key Studies

Table 3.1 summarizes the representative studies emphasizing on the MRI-to-X-ray registration and vertebral-feature methods most relevant to this project. The indicated metrics were achieved using various sets of data and protocols, which are provided for the convenience of the literature review; thus, these results cannot be directly compared to each other.

Table 3.1: Summary of relevant 3D–2D medical image registration studies.

Study	Task / modality	Method	Main reported finding	Limitation for this project	Relevance
Miao et al. (2016) [2]	3D CT/DRR to X-ray	Hierarchical CNN regression of pose parameters.	Direct prediction improves speed and capture range relative to intensity-based optimization in the evaluated tasks.	Not designed specifically for segmented spinal MRI or cross-modal MRI intensity.	Foundational direct-regression approach.
Jaganathan et al. (2021) [6]	X-ray to CT; rigid registration	Learned iterative updates with a point-to-plane correspondence solver.	Reports 0.60 ± 0.40 mm mean reprojection distance, 97% success, and approximately 8 s runtime.	Results depend on the study's training distribution and are not MRI-specific.	Useful hybrid learned/geometric baseline.
Ku et al. (2025) [10]	Pre-operative MRI to intra-operative fluoroscopy; femur/pelvis	MRI segmentation, synthetic CT generation, DRRs, and projection-based registration.	Demonstrates an MRI-to-fluoroscopy pipeline in simulation and cadaver studies for orthopaedic procedures.	Target anatomy is femur/pelvis rather than spine; uses an intermediate sCT representation.	Most direct modality-level precedent.

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Table 3.1: Summary of relevant 3D–2D medical image registration studies (continued).

Study	Task / modality	Method	Main reported finding	Limitation for this project	Relevance
Unberath et al. (2021) [1]	Review of machine learning for 2D/3D registration	Systematic review and perspective on learning, geometry, and verification.	Synthesizes methods and highlights domain shift, limited real-world data, and evaluation challenges.	Review rather than a single proposed registration algorithm.	Provides field-level context for the research gap.
Zhao et al. (2024) [12]	Spinal CT to X-ray	Two-step transformer with semantic attention and adaptive multi-dimensional loss.	Reports average mTRE of 0.96 mm on synthesized data and 2.32 mm on clinical data.	Uses CT rather than MRI; performance is linked to its specific data and geometry.	Strong precedent for segmentation/semantic spinal registration.
Jaganathan et al. (2023) [7]	CT projections to real X-ray	Self-supervised feature- and pixel-space domain adaptation.	Reports 1.83 ± 1.16 mm accuracy and 90.1% success on real X-ray images.	Domain adaptation targets X-ray/CT, not segmented MRI-to-X-ray alignment.	Informs synthetic-to-real robustness.

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Table 3.1: Summary of relevant 3D–2D medical image registration studies (continued).

Study	Task / modality	Method	Main reported finding	Limitation for this project	Relevance
Gopalakrishnan et al. (2024) [8]	Intra-operative X-ray to pre-operative CT volume	CNN pose initialization with differentiable X-ray rendering and test-time refinement.	Reports sub-millimetre accuracy on the evaluated surgical datasets.	Relies on a volume suitable for X-ray rendering and patient-specific refinement; not a segmentation-aware MRI study.	Supports geometry-aware refinement.
Qu et al. (2024) [11]	Spinal X-ray to CT	Anatomical feature extraction with graph and transformer-based modelling.	Demonstrates an anatomy-informed learning strategy for 2D/3D transformation estimation.	CT-to-X-ray setting and task-specific data; not MRI-to-X-ray.	Supports the use of vertebral anatomical cues.
Shen et al. (2026) [13]	Single-view lumbar X-ray to CT	Semantic vertebral-edge extraction to guide coarse-to-fine registration.	Reports approximately 1.5 mm mean in-plane translation error and around 10 s registration time on the evaluated clinical data.	Evaluated for CT-to-X-ray lumbar registration, with a limited patient cohort.	Recent semantic-edge-guided spinal method.

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Table 3.1: Summary of relevant 3D–2D medical image registration studies (continued).

Study	Task / modality	Method	Main reported finding	Limitation for this project	Relevance
Gopalakrishnan et al. (2026) [9]	Patient-specific X-ray to 3D volume registration	Self-supervised patient-specific network trained using physics-based simulated projections, with gradient-based optimization.	Reports adaptation in about five minutes and evaluation across anatomical structures and volumetric modalities.	Does not specifically study the added value of pre-existing segmented vertebral masks for spine MRI-to-X-ray.	Latest broad 3D-volume-to-X-ray precedent.
De Silva et al. (2017) [3]	Pre-operative MRI to intra-operative radiographs	Segmented MRI forward projection, gradient-orientation similarity, and CMA-ES.	Ten-patient clinical study; median projection-distance error 4.3 mm (IQR 2.6 mm), with no failures under the reported protocol.	Feature representation and optimizer are conventional; the study addresses vertebral-level localization.	Establishes prior clinical MRI-to-radiograph registration.
Kiran et al. (2024) [4]	Segmented MRI to simulated X-ray	Vertebral endplate feature extraction, forward projection, binary image matching, and CMA-ES.	Simulated-view evaluation; reports about 95% success for single-view tests and around 60 s mean runtime.	Main experiments use simulated X-ray views rather than a full real-X-ray evaluation.	Direct precedent for simulated X-rays and feature-based matching.

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Table 3.1: Summary of relevant 3D–2D medical image registration studies (continued).

Study	Task / modality	Method	Main reported finding	Limitation for this project	Relevance
Kiran et al. (2026) [5]	Pre-operative MRI to intra-operative X-ray; pedicle trajectory mapping	Endplate feature extraction, forward projection, binary matching, and CMA-ES; includes downstream trajectory evaluation.	Reports a 3 mm MRI slice thickness and PDE below 5 mm and TRE below 2.5 mm in its evaluated setting.	Closely related prior work; dataset provenance and the present study's distinct contribution must be stated.	Most directly related real-X-ray precedent.

Chapter 4

Research Gap

The literature already contains both clinical MRI-to-X-ray spinal registration and feature-based approaches. De Silva et al. aligned segmented pre-operative MRI to intra-operative radiographs by forward projection and gradient-orientation similarity[3]. Kiran et al. later produced simulated X-ray views from segmented MRI and estimated vertebral pose from endplate features, binary-image matching, and CMA-ES[4]. and a subsequent study combined endplate-based registration with trajectory mapping using MRI (3 mm slice thickness) and intra-operative X-rays[5]. The proposed research therefore does not claim novelty for MRI-to-X-ray registration, simulated projection generation, or feature-based registration.

The research question is if learned feature extraction or description improves an anatomy-aware registration pipeline for pre-operative spinal MRI and intra-operative X-ray. To attribute a measured difference to the feature representation it is necessary to base a baseline and learning-based method on the same cases, projection geometry, pose initialization, downstream optimization, and evaluation protocol wherever possible. Results on simulated projections, where the generating transformation is known, need to be reported separately from results on real intra-operative X-rays, where reference measurements may be incomplete.

The study will contrast a standard vertebral-feature baseline with a learned feature representation using simulated projections from MRI-derived anatomy and the available real X-rays. Evaluation will focus on registration error, computation time, and sensitivity to controlled changes in the initial pose. The model's complexity and the validation protocol will be determined by the number of independent cases and the availability of projection geometry, segmentation masks, and reference landmarks. Any contribution claims should pertain to the measured effect of the learned representation on this dataset, not present segmentation, feature matching, or simulated X-rays as novel for spinal MRI-to-X-ray registration.

Chapter 5

Objectives

The proposed research has the following objectives:

1. To critically evaluate approaches based on deep learning and conventional feature-based algorithms to rigid 3D–2D spinal MRI-to-X-ray registration, with a focus on anatomical feature extraction, simulated projection, and geometric pose estimation.

2. To devise a feature-based registration pipeline that extracts vertebral features from pre-operative MRI, simulate 2D projections with an appropriate projection geometry and match them to features extracted from intra-operative X-ray images.

3. To assess if using learning-based features improves the registration compared to a conventional vertebral-feature baseline while maintaining other registration conditions as similar as possible.

4. To evaluate registration accuracy, computational efficiency, and robustness to controlled initial-pose variations on simulated projection data with known transformations and, when available, real X-ray images with appropriate reference measurements.

Chapter 6

Dataset Description

The input data comprises pre-operative spinal MRI with a 3 mm slice thickness and a two-dimensional intra-operative X-ray image. The MRI gives the 3D anatomical structure of the patient's spine whereas the X-ray image is the projection onto which the MRI voxels should be matched. It should be noted that the method does not require a synthesis of a CT volume.

Preparation will begin with checks of the MRI and X-ray metadata, identification of the corresponding vertebral region and preservation of the spatial coordinates needed for projection and registration. Available vertebral masks, endplate contours, landmarks or other reliable anatomical annotations will be used where possible. The 3 mm slice thickness must be reflected when converting voxel indices to physical coordinates. Any resampling will be documented and will not be described as restoring through-plane detail that was not captured by the original scan.

Simulated views will be generated by projecting onto the available or estimated projection geometry MRI-derived anatomical features. As MRI intensities are not related to X-ray attenuation coefficients, these views will represent either geometry- or feature-based projections unless a validated mapping from MRI intensity values to attenuation coefficients is available. They are meant to provide examples with known transformations rather than attempt to simulate the image formation process of a radiograph from raw MRI intensities. Actual intra-operative X-rays will be a separate test set to evaluate performance beyond simulated data.

Beforehand experimentation is done, a database inventory of independent cases, vertebral levels, X-ray views, image dimensions, MRI orientation and voxel spacing, projection calibration, annotations, and reference measurements, will be taken. Since that information is not known yet in this description, it would not be assumed. The database provenance should also be compared with published studies that used 3 mm MRI, and intra-operative X-rays, so there can be disclosure if they overlap, and distinction between this present work, and the former one.

Chapter 7

Evaluation Metrics

Chapter 8

Proposed System Block Diagram

Figure 8.1 summarizes the proposed hybrid workflow. Vertebral features are extracted from the MRI and forward-projected through a projection model to generate surrogate 2D feature views, which are jointly paired with anatomical features extracted from the real intra-operative X-ray. The same pair of inputs are used to compare a conventional feature-based baseline with a learning based representation. The diagram is conceptual; the encoder and optimizer would be selected after the availability of annotations, number of independent cases, and projection geometry has been verified.

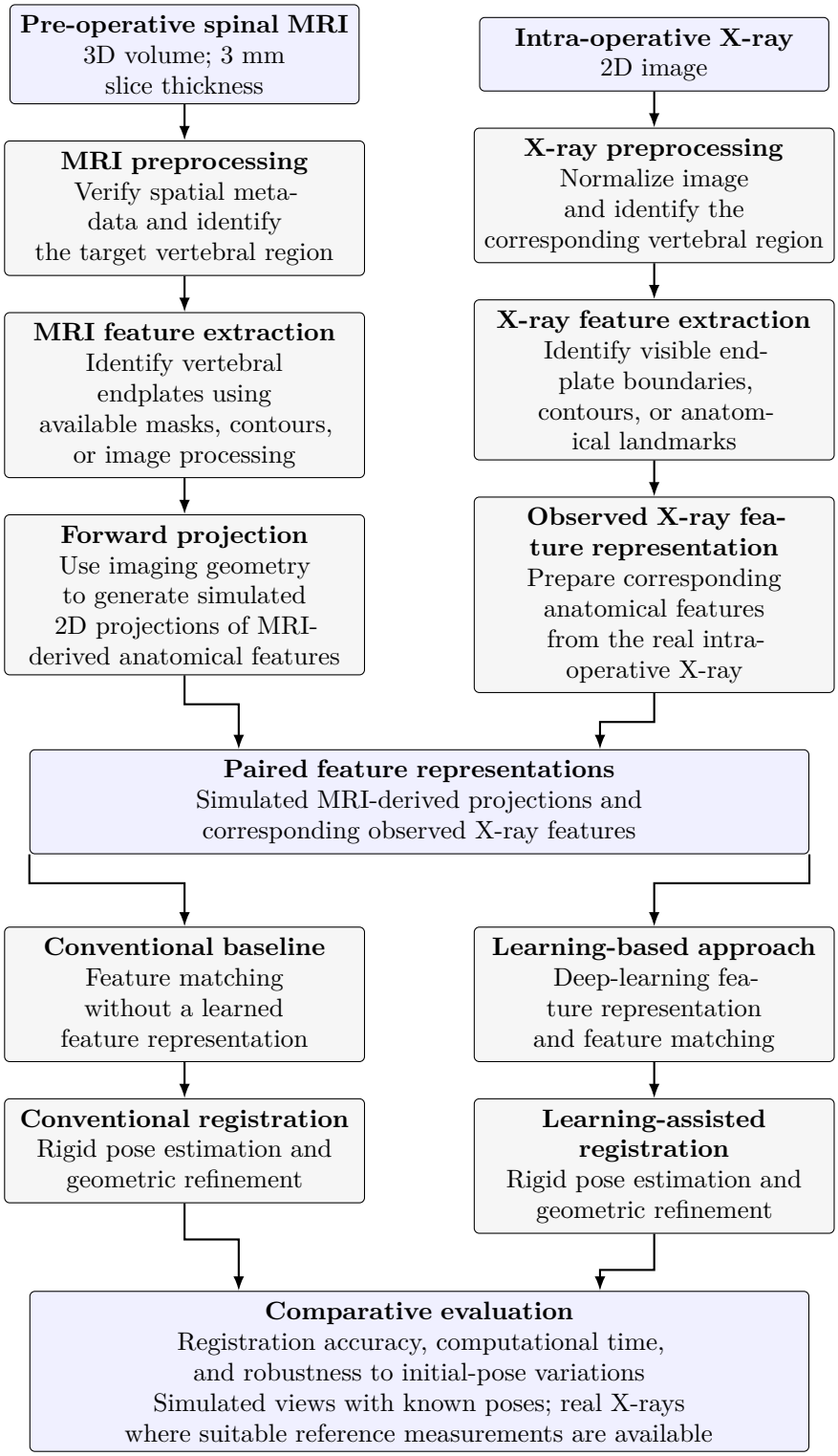


Figure 8.1: Conceptual workflow comparing conventional and learning-based feature representations for rigid 3D–2D spinal MRI-to-X-ray registration.

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